





STUDENT FULL NAMES: _____ Date__/10/2025

Section A: Multiple Choice (20%)

1. Choose the correct answer. ***(2 marks each)***

1. Which of the following best defines machine learning? a) A program that follows fixed instructions b) A subset of AI that learns from data without explicit programming c) A database for storing large data d) A manual process for analyzing information	2. The first step in the Machine Learning Life Cycle is: a) Data preparation b) Problem definition c) Model deployment d) Model training
3. Which tool is most widely used for data manipulation and cleaning in Python? a) NumPy b) Pandas c) Matplotlib d) TensorFlow	4. A dataset with labels provided is used in: a) Unsupervised learning b) Reinforcement learning c) Supervised learning d) Semi-supervised learning
5. Which of the following represents the 6 V's of Big Data? a) Value, Variety, Validity, Variables, Version, Velocity b) Volume, Variety, Velocity, Veracity, Variability, Value c) Vision, Volume, Variation, View, Vector, Variance d) None of the above	6. During model evaluation, which of the following is used for classification accuracy? a) RMSE b) Precision & Recall c) Mean Squared Error d) Standard deviation
7. Which IDE is cloud-based and allows GPU/TPU use for ML training? a) PyCharm b) Spyder c) Google Colab d) Visual Studio Code	8. Which ML algorithm is best suited for clustering? a) K-Means b) Linear Regression c) Decision Trees d) Random Forest
9. Which of the following is a disadvantage of ML? a) Continuous improvement b) High computational cost c) Wide range of applicability d) Improved decision making	10. Which of these datasets is built-in with Scikit-learn? a) CIFAR-10 b) IMDB reviews c) Iris dataset d) MS COCO

2. Discuss what you have learned in this course: "Machine learning" – ***(10 marks)***

Section B: Matching (10%)

Match the items in Column A with their correct description in Column B. ***(1 mark each)***

Column A:	Column B:
1. Supervised Learning	A. Groups unlabeled data into clusters
2. Unsupervised Learning	B. Learns from interaction with environment
3. Reinforcement Learning	C. Uses both labeled & unlabeled data
4. Semi-supervised Learning	D. Predicts outcomes using labeled datasets

Section C: Fill in the Blanks (10%)

(2 mark each)

1. The programming language most widely used in ML is _____.
2. _____ is the library used for numerical computation in Python.
3. In supervised learning, the data must be _____.
4. A _____ is a large, centralized storage system for data from multiple sources.
5. The ML framework developed by Google is called _____.

Section D: Open Questions (50%)

11. Explain the difference between Artificial Intelligence, Machine Learning, and Deep Learning. Give at least 2 differences for each pair. ***(8 marks)***

12. Practical Task: Describe the main steps involved in Installation Python in your lab computer and in verifying the installation. ***(6 marks)***

13. You are provided with raw data containing missing values and duplicates.

    a) Explain the steps of data cleaning. ***(2 marks)***

    b) Write short Python code to remove duplicates using Pandas. ***(8 marks)***

14. An online store wants to recommend products to customers based on past purchases.

    a) Which ML type would you use and why? ***(2 marks)***

    b) Suggest one ML algorithm suitable for this task. ***(8 marks)***

15. Discuss two advantages and two disadvantages of using Big Data in healthcare decision making. ***(6 marks)***

16. Design a mini-project outline (3 - 4 steps) to collect, preprocess, and visualize a dataset (example: predicting student performance or detecting spam emails). ***(10 marks)***